1 Solution — GIAM 1.4 Problem 6

1.1 Problem

In hexadecimal (base-16) the sixteen “digits” are 0–9 together with A–F for 10 through 15. Write the next 10 hexadecimal numbers after AB, and the next 10 after FA.

1.2 Hexadecimal Digits

For reference, the sixteen hex digits and their decimal values:

hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Table 1.1

Counting works like any base: increase the units digit until it passes F $(= 15)$, then roll it over to 0 and carry 1 into the next place.

1.3 The Next 10 After AB

$$\boxed{AC,\ \ AD,\ \ AE,\ \ AF,\ \ B0,\ \ B1,\ \ B2,\ \ B3,\ \ B4,\ \ B5.}$$

The units digit climbs $B \to C \to D \to E \to F$, giving AC, AD, AE, AF. At AF the units digit is maxed out, so the next step rolls it to 0 and carries: $A \to B$, producing $B0$. Then $B1, B2, \ldots, B5$.

1.4 The Next 10 After FA

$$\boxed{FB,\ \ FC,\ \ FD,\ \ FE,\ \ FF,\ \ 100,\ \ 101,\ \ 102,\ \ 103,\ \ 104.}$$

The units digit climbs $A \to B \to C \to D \to E \to F$, giving FB, FC, FD, FE, FF. Now FF has *both* digits maxed out, so the next step is a full rollover — both digits go to 0 and a new leading digit appears: $FF + 1 = 100_{16}$. Then $101, 102, 103, 104$.

1.5 Counting Through the Carries

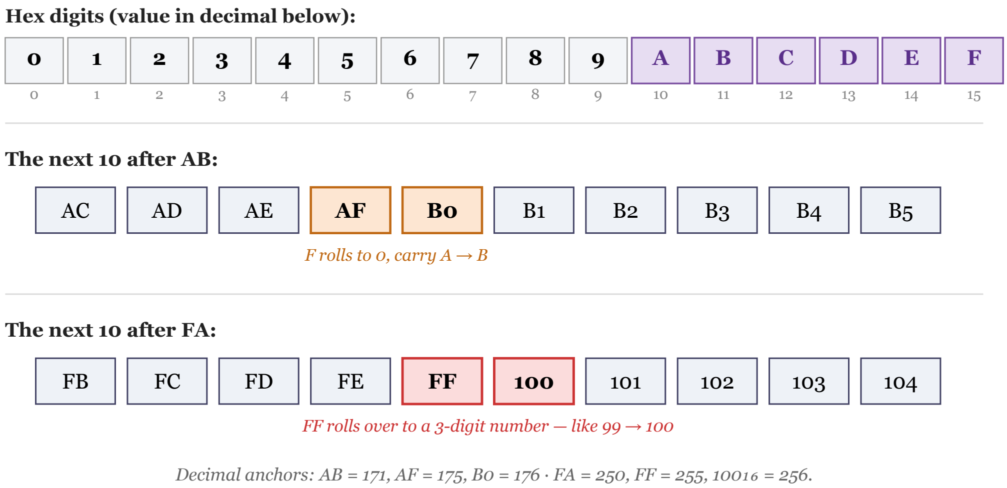

Figure 1.1: Top: the hex digits 0–F with A–F (the letter digits) highlighted. Middle: after AB, the units digit runs to AF then carries to B0 (F→0, A→B). Bottom: after FA, the digits run to FF then roll over to the three-digit 100_{16} — like 99 → 100. Decimal anchors: AB = 171, AF = 175, B0 = 176; FA = 250, FF = 255, 100_{16} = 256.

1.6 Remark — Two Kinds of Rollover

The two sequences showcase the *single* carry and the *full* carry:

- $AF \to B0$ is a single carry: only the units digit is maxed, so it resets to 0 and bumps the next digit up by one ($A \to B$). This is the hex twin of $19 \to 20$ in decimal.
- $FF \to 100$ is a cascading carry: *every* digit is maxed (F), so they all reset to 0 and a fresh leading 1 is born. This is the hex twin of $99 \to 100$ — and exactly the $777_8 \to 1000_8$ rollover from Problem 3 and $7 \to 10_8$, $9 \to 10$ generally. In *any* base, a string of maximal digits rolls over to "1 followed by zeros."

In decimal terms, $FF_{16} = 255 = 16^2 - 1$ and $100_{16} = 256 = 16^2$ — one more than the maxed two-digit number, opening a new power-of-16 place. (Just as $99 = 10^2 - 1$ rolls to $100 = 10^2$.)

1.7 Remark — Decimal Cross-Check

Converting to decimal confirms the counts are consecutive:

hex	$=$	**decimal**
AB	$10 \cdot 16 + 11$	171
AF	$10 \cdot 16 + 15$	175
$B0$	$11 \cdot 16 + 0$	176
FA	$15 \cdot 16 + 10$	250
FF	$15 \cdot 16 + 15$	255
100_{16}	$1 \cdot 16^2$	256

Table 1.2

So “next 10 after AB” is 172 through 181, and “next 10 after FA” is 251 through 260 — each list a run of ten consecutive integers, exactly as counting should give.

1.8 Remark — Hexadecimal Is Binary, Four Bits at a Time

Just as octal packs binary three bits at a time (Problem 3), hexadecimal packs it **four** bits at a time, because $16 = 2^4$. Each hex digit corresponds to a 4-bit block (0000 through 1111):

$$AB_{16} = \underbrace{1010}_{A}\,\underbrace{1011}_{B}{}_2 = 10101011_2 = 171, \qquad FF_{16} = \underbrace{1111}_{F}\,\underbrace{1111}_{F}{}_2 = 11111111_2 = 2^8 -$$

This is why hexadecimal is *the* notation of computing: a **byte** is 8 bits $=$ exactly two hex digits, so any byte is written with two hex characters, and the largest byte value is

$FF = 255 = 2^8 - 1$. The full rollover $FF \to 100_{16}$ is the moment a value outgrows a single byte — the hexadecimal echo of $2^8 - 1 \to 2^8$.

1.9 Remark — The General Rollover Rule

Across Problems 3–6 the same odometer principle recurs in every base b: count up the units digit through $b-1$, then carry. A maximal string $(b{-}1)(b{-}1)\cdots(b{-}1)$ of length k always rolls to $1\underbrace{00\cdots0}_{k}$, the value b^k. We have now seen it in four bases:

$$9 \to 10\ (b{=}10), \qquad 1_2\text{-runs} : 1 \to 10_2\ (b{=}2), \qquad 7 \to 10_8\ (b{=}8), \qquad F \to 10_{16}\ (b{=}16),$$

and the two-digit versions $99 \to 100$, $11_2 \to 100_2$, $77_8 \to 100_8$, $FF_{16} \to 100_{16}$. Hexadecimal feels exotic only because of the unfamiliar letter-digits A–F; the *mechanics* of counting are identical to decimal, which is the real takeaway of the exercise.